Gaussian Image Steganography via Parameter-Domain Keyed Embeddings

“Hide information in Gaussian parameters — not pixels”

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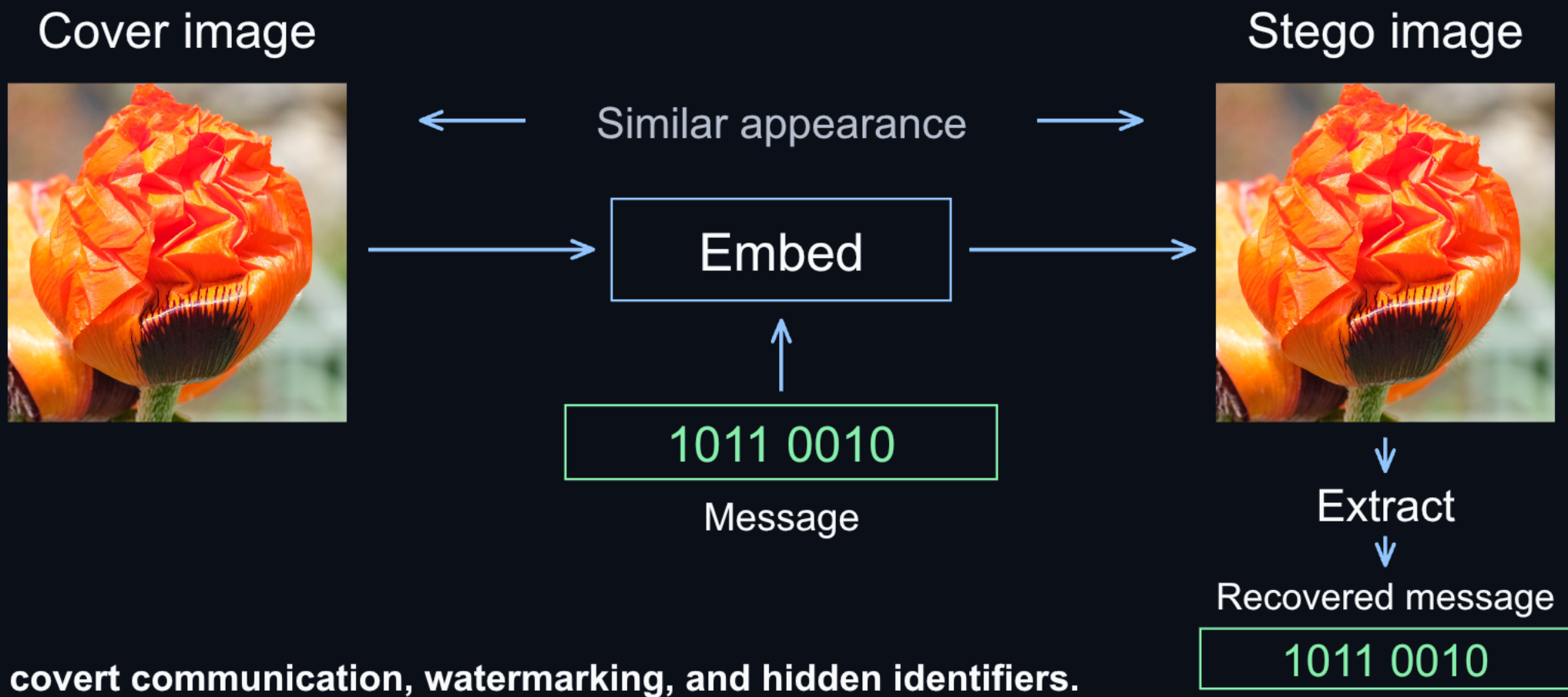
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What is steganography?

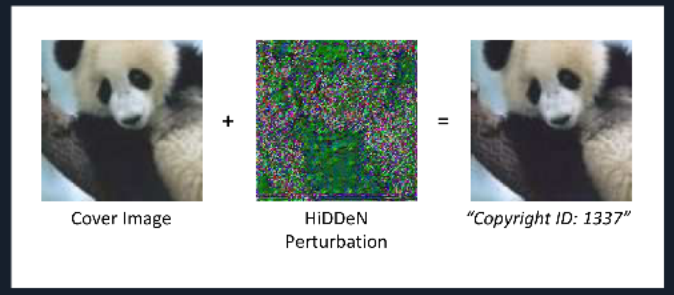
Steganography hides a message in a carrier while aiming to conceal its presence.



Uses: covert communication, watermarking, and hidden identifiers.

Where can hidden information live?

Image pixels



HiDDeN

Small changes to image pixels carry the message.

Learned latent representations



RoSteALS

The message is added to an autoencoder's latent code.

3D Gaussian representations



GaussianMarker / 3D-GSW

GaussianMarker reads parameters and renders. 3D-GSW reads rendered views.

HiDDeN, ECCV 2018 · RoSteALS, CVPRW 2023 · GaussianMarker, NeurIPS 2024 · 3D-GSW, CVPR 2025.

One image represented by 2D Gaussian parameters

Fitted clean parameters

i	μ	s	θ	c	α
1	•	⬭	⬭	●	●
2	•	⬭	⬭	●	●
3	•	⬭	⬭	●	●
...	⋮	⋮	⋮	⋮	⋮
N	•	⬭	⬭	●	●

Each row describes one Gaussian.

One image, represented by a set of 2D Gaussians

Position (μ) · Shape (s)
Orientation (θ) · Color (c)
Opacity (α)

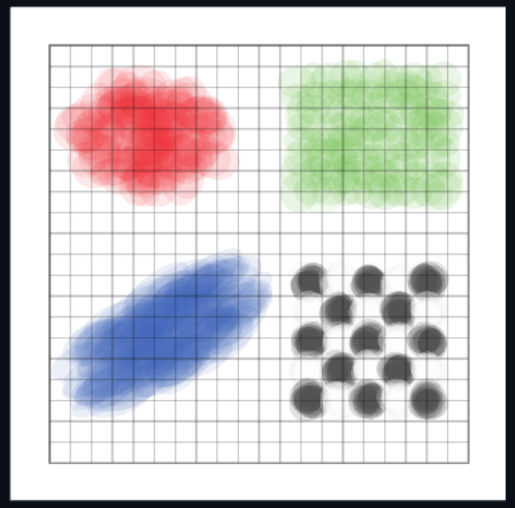
Can selected parameter changes carry a hidden message?

Keep the rendered image close to the clean fit.

An image can be represented by 2D Gaussians



Position | Shape and orientation | Color | Opacity



Overlapping soft, colored primitives form an image.

We send the Gaussian parameters to the receiver, who can use them to render the image.

Fitting Gaussians to the target image

Fitting adjusts the Gaussian parameters to approximate the target.

Target Image



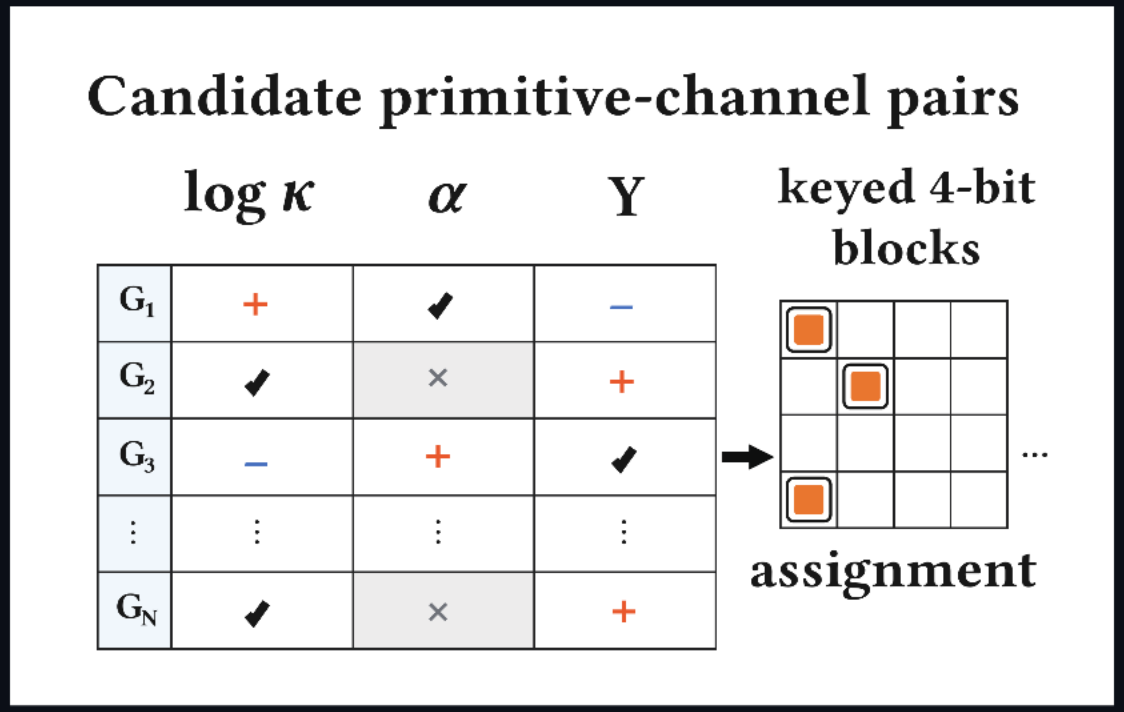
Gaussian Parameter Updates



This is the clean fit, before message embedding.

Can we add a message while keeping the rendering nearly unchanged?

Which parameter changes should carry the message?



01 — Low visual impact

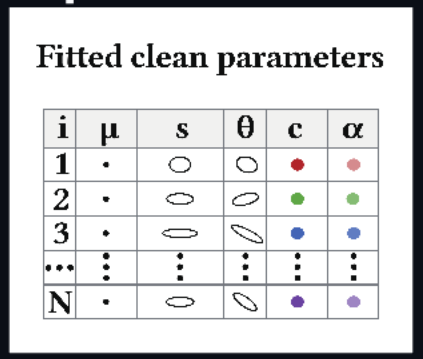
02 — Reliable detection

03 — Correct message

We need edits that satisfy all three requirements.

Choose the edits, then optimize the parameters

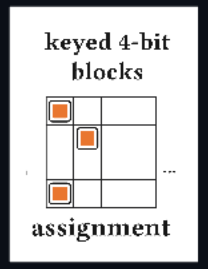
Clean parameters



Message + Secret key

Keyed assignment

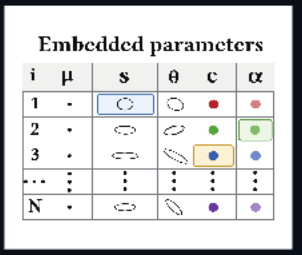
Assignment map



Selected edits

Fine-tuning

Embedded parameters



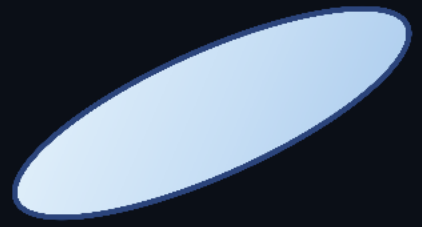
Assignment chooses the edits. Fine-tuning embeds the message.

Three parameters we can change

Elongation — log κ

How stretched
the Gaussian is

+Δ



More stretched

−Δ



More rounded

Opacity — α

How strongly it contributes
to the image

+Δ



Stronger contribution

−Δ



Weaker contribution

Brightness — Y

How bright
its color is

+Δ



Brighter

−Δ



Darker

**One edit specifies a Gaussian, a parameter,
and an increase or decrease.**

Choose valid edits with lower cost

Candidate edits

E_1 Increase elongation of Gaussian 1

E_2 Increase opacity of Gaussian 3

E_3 Decrease brightness of Gaussian 1

An edit set uses several candidate edits together.

Example edit set: E_1 and E_2

Select an edit set

1. **Keep sets that encode the target message.**

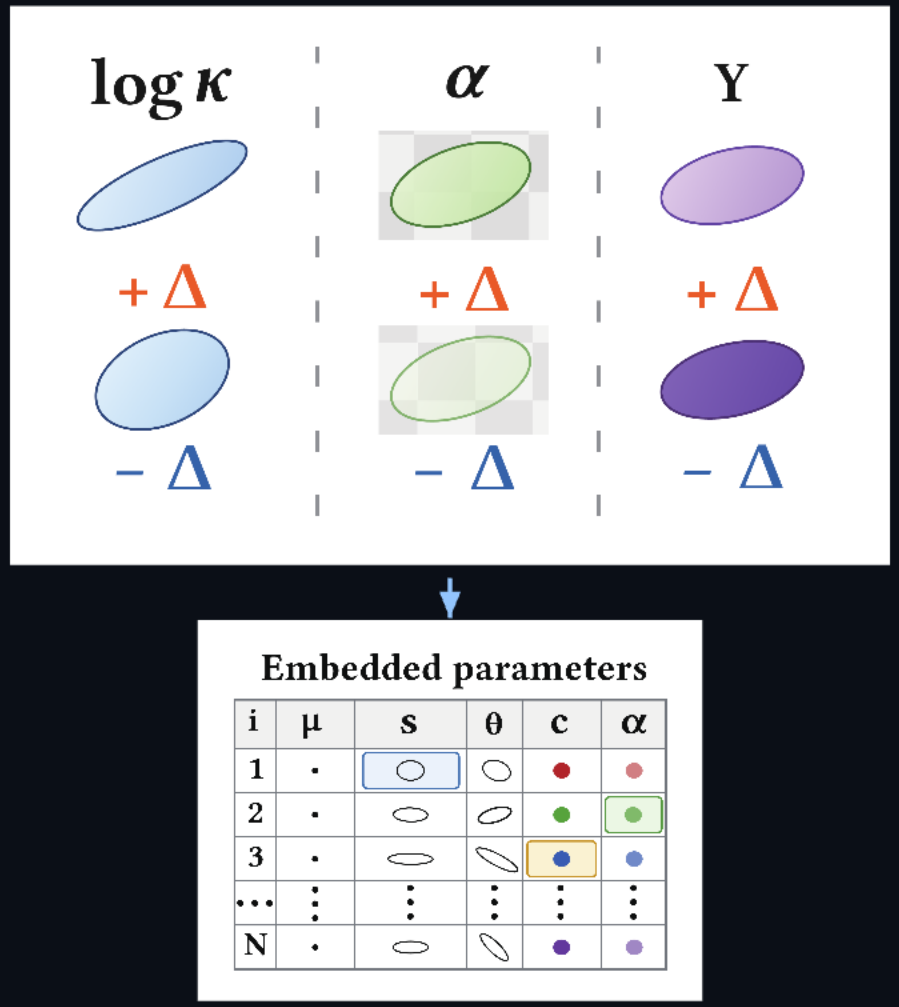
The key defines the candidate edits and their mapping to message bits.

2. **Among valid sets, prefer lower total cost.**

Cost estimates visual and parameter changes.

Region and coverage constraints also apply.

Fine-tuning refines the selected parameter edits



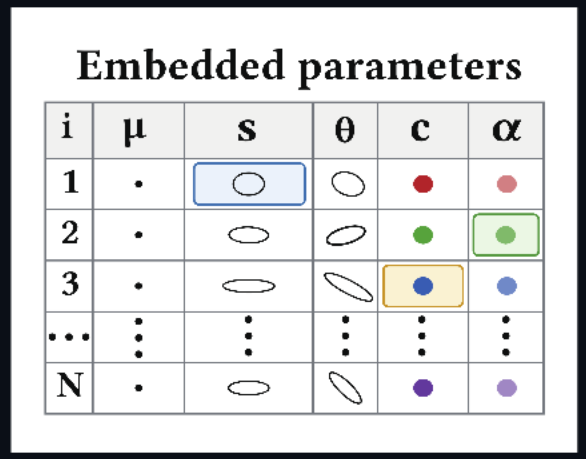
01 Make the intended changes readable.

02 Preserve the rendered appearance.

03 Limit changes to the parameter distribution.

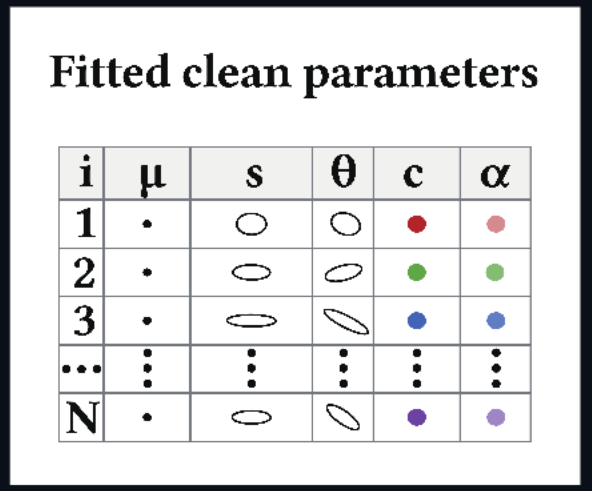
Read the message from parameter changes

Embedded parameters

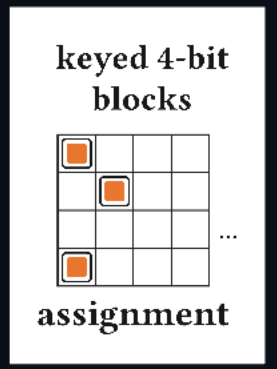


compare

Clean parameters before embedding



Assignment map



Which edits to check
How to read their changes
How to combine the decisions

Measure parameter changes → **Convert changes into 0/1 decisions** → **Recover four-bit message blocks**

Fixed 8-bit message: 256 Gaussians per synthetic fit / 4,096 per natural image.

The clean and embedded images look similar

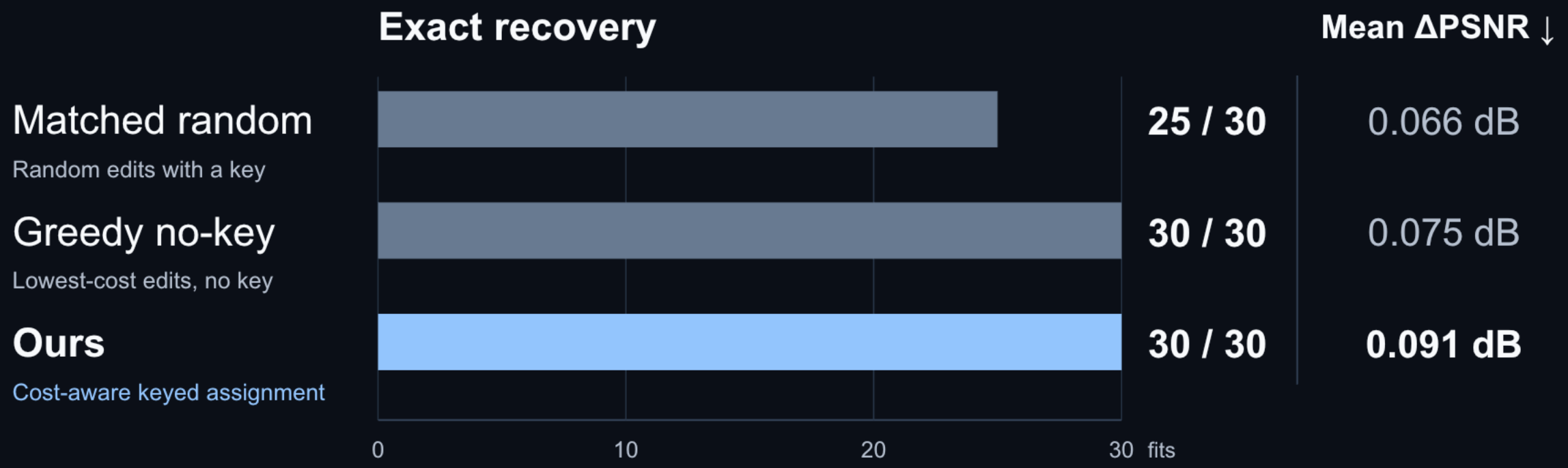
Source image	Clean Gaussian fit	Embedded Gaussian fit
		
		

Perceptually similar results

$\Delta PSNR = \max(0, PSNR_{clean} - PSNR_{embedded})$

Cost-aware selection improves recovery over random selection

30 synthetic fits / 256 Gaussians / fixed 8-bit message



Both methods recover every message in this test.

Greedy no-key selects low-cost edits without a key. It does not address key-specific decoding.

Our method uses a key to define candidate edits and their mapping to message bits.

Correct-key BER: 0.000. Mean wrong-key BER: 0.543 (chance level: 0.5).

The fixed message is recovered from all 112 test images

112 held-out images / 256 × 256 / 4,096 Gaussians / fixed 8-bit message

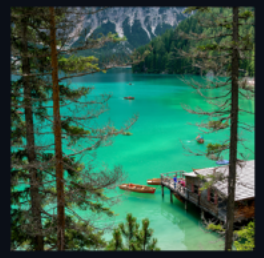
Examples from the held-out input set


0801


0803


0805


0806


0807


0808


0809


0810

112 / 112

Exact message recovery

Correct-key BER: 0.000

0.009 dB

Mean ΔPSNR

0.520

Mean wrong-key BER

Chance BER = 0.5

Thank you

Paper / Supplementary material / Code / Slides






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Citation

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